

8.7 Completing the Square Algebraically

Lesson Introduction

 Benchmarks	MA.912.AR.3.6 Given an expression or equation representing a quadratic function, determine the vertex and zeros and interpret them in terms of a real-world context. MA.912.AR.1.2 Rearrange equations or formulas to isolate a quantity of interest.		 Learning Targets	- I can complete the square for a quadratic expression algebraically. - I can rewrite a quadratic equation in vertex form by completing the square.	
 Benchmark Coherence	Building On N/A		Working Toward N/A		
 Essential Questions	- How can I relate completing the square with algebra tiles to an algebraic method? - What are the steps to completing the square algebraically?				
 Lesson Narrative	In this lesson, students move from the conceptual understanding of completing the square to completing the square algebraically. Students will transition into an algebraic approach by learning the steps necessary to complete the square for a given quadratic equation.				
 Component Summary	8.7.1 Warm-Up (~5 min.)	Analyze a square pattern to determine the calculation and number of squares (<i>SE p. 43</i>)	8.7.3 Guided Practice (15 min.)	Engage in a structure practice of completing the square algebraically (<i>SE p. 46</i>)	
	8.7.2 Collaboration (20 min.)	Collaborate to connect using algebra tiles to complete the square to being able to do it algebraically (<i>SE p. 43</i>)	8.7.4 Wrap-Up (~5 min.)	Complete a Ticket Out the Door that assesses rewriting into vertex form by completing the square and identifying key features (<i>SE p. 47</i>)	
 Resources	Glossary		Materials		Additional Preparation
	<u>Key Terms:</u> N/A <u>Additional Terms:</u> - Vertex Form - Perfect Square Trinomial		(Optional) Chart Paper & Markers (Optional) Algebra Tiles		Work through the 8.7.2 Collaboration activity prior to teaching. Highlight areas that may cause pacing concerns based on student performance and determine instructional supports you may want to place.

 Teacher Tips	Common Misconceptions	Things to Consider
	- Students may use the wrong sign when writing the binomials (see table in 8.7.2 Collaboration). - Students may struggle with determining the value that makes a perfect square trinomial. For example, students may multiply by 2 rather than squaring the number.	Avoid students using the table in 8.7.2 Collaboration as a list of steps to memorize when solving algebraically. Utilize routines, like Three Reads and talk moves like revoicing or rewording to encourage conceptual understanding.
	Literacy and Language Routines	Mathematical Thinking and Reasoning (MTR) Standard
	Three Reads Consider utilizing the Three Reads language routine structure when students are reviewing the table provided in 8.7.2 Collaboration. Students may initially see the amount of text as a barrier to learning. Three Reads provides scaffolding for students to find more intentional entry-points to the information.	MA.K12.MTR.5.1: Patterns and Structure Students will be connecting what they learned about the concept and process of completing the square using algebra tiles to the new concept of completing the square algebraically. Encourage students to make connections throughout 8.7.2 Collaboration, transferring pictorial representations to abstract concepts of solving algebraically.
 Differentiate Instruction	During 8.7.2 Collaboration, consider creating a Foldable or other form of note-taking visual aid to assist learners with a guided reference for how to complete the square algebraically. Avoid instruction that highlights steps to memorize and emphasize sensemaking when constructing scaffolding strategies. - Anchor Chart with samples illustrated using colors, shapes, or symbols - Foldable that includes the “why” behind each learning opportunity - Voice or video recordings as a “study expert” to share the process or what they learned	
Lesson Notes		

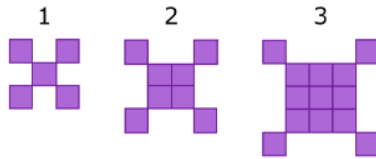
8.7.1 Warm-Up: Square Pattern

Component Overview: Students analyze a square pattern to determine the calculation and number of squares.



8.7.1 Warm-Up: Square Pattern

Three steps of a visual pattern are shown.



1. Use the pattern to complete the table.

n	Calculation	# of squares
1	$1^2 + 4$	5
2	$2^2 + 4$	8
3	$3^2 + 4$	13
4	$4^2 + 4$	20
43	$43^2 + 4$	1853
n	$n^2 + 4$	$n^2 + 4$



Consider having students share out their methods and processes to determine the pattern. Allow students to defend their pattern. Encourage different paths and thinking.



For a challenge, consider removing the calculation column of the table in order to remove some of the structures and stick strictly to asking students for the total amount of squares in the 43rd step in the pattern.

8.7.2 Collaboration: Completing the Square Algebraically

Component Overview: Students will collaborate to connect using algebra tiles to complete the square to being able to do it algebraically.

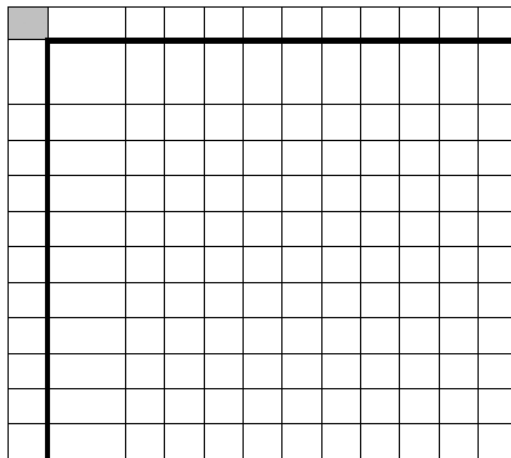


8.7.2 Collaboration: Completing the Square Algebraically

For each expression, the first step to completing the square is shown.

Expression A	Expression B	Expression C
$6x^2 - 24x + 15$	$x^2 - 24x + 15$	$4x^2 - 24x + 15$
$6(x^2 - 4x) + 15$	$(x^2 - 24x) + 15$	$4(x^2 - 6x) + 15$

- Discuss with your partner how to determine the value to add to the expression inside the parentheses that results in a perfect square trinomial. Use the algebra tiles diagram provided if sketching a model will help you.



Answers will vary. To determine the value to add to the expression, take half of the value of the coefficient of the x -term, then square it.



Students are working in pairs. Consider using prompting questions while circulating the room to extend student thinking where necessary.

- What is the quadratic form the original expressions are written in?
- What key features can be revealed when quadratic equations are written in this form?
- Why might these expressions need to be rewritten into vertex form?



Some students may prefer sketching models over physical algebra tiles, although they have only one mat in their workbook. Consider providing small copies of mats for students to utilize during Collaboration.

8.7.2 Collaboration: Completing the Square Algebraically (continued)

2. Use your process to complete each perfect square trinomial.

Expression A	$6(x^2 - 4x + 4) + 15$
Expression B	$(x^2 - 24x + 144) + 15$
Expression C	$4(x^2 - 6x + 9) + 15$

3. In Expression A, the value that was added to create the perfect square trinomial is being multiplied by 6, the coefficient of the perfect square trinomial. The value that should be subtracted from 15 to maintain equivalence is 24.
4. In Expression B, the value that was added to create the perfect square trinomial is being multiplied by 1, so the value that should be subtracted from 15 to maintain equivalence is 144.
5. In Expression C, the value that was added to create the perfect square trinomial is being multiplied by 4, the coefficient of the perfect square trinomial. The value that should be subtracted from 15 to maintain equivalence is 36.



Consider using a jigsaw for differentiating content based on your students. Each of the three groups can be assigned to Expression A, B, or C. Have students work together in groups to complete questions 2, 3, and 6 with their assigned expression before finding a student from a different group and sharing their strategies and answers.



Consider inquiry to extend student thinking in small groups either as you circulate the room or as prompts on cards for students to answer.

- Expression A: Did anyone get -11 ? If so, what did you do about the coefficient of 6?
- Expression B: Did anyone get 159? If so, did you add or subtract 144? Why?
- Expression C: Did anyone get 6? Did you subtract 9? If so, and if I told you that you had to use the coefficient 4, how would that change your answer?
- Synthesis: What do you notice about the process of completing the square using algebra tiles and the statement in questions 3-5?

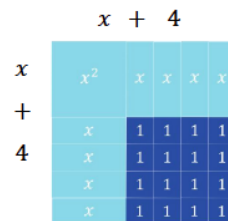
8.7.2 Collaboration: Completing the Square Algebraically (continued)

6. Factor each perfect square trinomial and rewrite as a binomial squared. Also, enter the value determined from the steps described above, including the appropriate sign (+ or -).

Expression A	$6(x - 2)^2 + 15 - 24$
Expression B	$(x - 12)^2 + 15 - 144$
Expression C	$4(x - 3)^2 + 15 - 36$

Rewriting the resulting perfect square trinomial as a binomial squared is the next step in the process as it represents the factored form of a perfect square trinomial. The algebra tiles are a visual representation of this perfect square.

$$x^2 + 8x + 16 = (x + 4)^2$$



The process of completing the square is used to rewrite quadratic expressions, equations, or functions from standard form, $ax^2 + bx + c$, to vertex form, $a(x - h)^2 + k$.

This process is helpful when trying to find key features that are not easy to determine using standard form.



Consider providing a station or projection located centrally for students to opt-in for additional scaffolding as they collaborate. Students can utilize the tiles as scaffolding to transition from pictorial representation to conceptual understanding.



Would a negative c term ever be possible if the b term is negative? Give an example.

8.7.2 Collaboration: Completing the Square Algebraically (continued)

7. While learning the process of completing the square with algebra tiles, the leading coefficients were all positive. The table shown gives a quadratic equation in standard form that has negative coefficients.

Complete the table to complete the square without using algebra tiles.

Process	Equation
Original equation	$y = -3x^2 - 24x + 11$
a. Factor out the common factor of the first two terms.	$y = -3(x^2 + 8x) + 11$
b. Discuss with your partner why there is a plus sign instead of a minus sign in the parentheses. $(x^2 + \text{ } x)$	
c. Complete the statement. There is a plus sign instead of a minus sign in the parentheses because ... <i>Answers will vary. Because the common factor is a -3, so $-24 \div -3$ is a positive.</i>	
d. Determine the value that makes a perfect square trinomial in the parentheses.	$y = -3(x^2 + 8x + 16) + 11$
e. Then, to maintain equivalence, multiply the factor by the value . Add the opposite of that value .	$y = -3(x^2 + 8x + 16) + 11 + 48$



Notice the sentence frame provided for making sense of signs in the trinomial. Highlight this moment as a learning opportunity, eliciting student responses whole group to clarify misconceptions.

8.7.2 Collaboration: Completing the Square Algebraically (continued)

Process	Equation
f. Discuss with your partner why, after finding the product of the factor and the value added to make a perfect square trinomial, the opposite of this product is added to the expression. Summarize your discussion. <i>Answers will vary. You add the opposite of the value so that equivalence is maintained.</i>	
g. Rewrite the perfect square trinomial represented in the parentheses as a binomial squared.	$y = -3(x + 4)^2 + 11 + 48$
h. Combine like terms.	$y = -3(x + 4)^2 + 59$



Provide opportunities for students to make sense of information. Utilize routines like Think-Pair-Share, Three Reads, or Collect and Display to ensure students are not just memorizing steps but instead moving from algebra tiles to conceptual understanding.



Consider creating an Anchor Chart that represents the information of the steps in completing the square algebraically. If enough space is provided, consider drawing the algebra tile version as well so students can have a visual reference of that connection.

8.7.3 Guided Practice: Completing the Square Algebraically

Component Overview: Students will engage in a structure practice of completing the square algebraically to find key features in vertex form.



8.7.3 Guided Practice: Completing the Square Algebraically

Rewrite each equation in vertex form by completing the square. Then, give the key features revealed by both equations.

1. $y = x^2 + 22x - 53$
 $y = (x + 11)^2 - 174$

Key Features					
x-intercept		vertex	$(-11, -174)$	domain	R
x-intercept		increasing	$x > -11$	range	$y \geq -174$
y-intercept	$(0, -53)$	decreasing	$x < -11$	opens	up
left end behavior	$x \rightarrow -\infty, y \rightarrow \infty$				
right end behavior	$x \rightarrow \infty, y \rightarrow \infty$				

2. $f(x) = -8x^2 + 80x - 71$
 $f(x) = -8(x - 5)^2 + 129$

Key Features					
x-intercept		vertex	$(5, 129)$	domain	R
x-intercept		increasing	$x < 5$	range	$y \leq 129$
y-intercept	$(0, -71)$	decreasing	$x > 5$	opens	Down
left end behavior	$x \rightarrow -\infty, y \rightarrow -\infty$				
right end behavior	$x \rightarrow \infty, y \rightarrow -\infty$				



Prompt students to use the scaffolding from 8.7.2 Collaboration for additional support, or pull a small group based on your observations during 8.7.2 Collaboration.



Consider the following questions for students who are stuck or need additional support:

- What key features are only revealed when an equation is written in vertex form?
- How can you determine the end behavior in this form?
- Why are the intercepts not revealed from this form?
- What form of a quadratic equation reveals the intercepts?



For questions 1-2, students should only fill in the key features that are revealed. Students should not have to provide key features that cannot be found using either form.



If students finish early, have students create an equation in standard form and trade with a partner to try and write in vertex form. Students may need to create one in factored form first and then multiply to create it in standard form.

8.7.4 Wrap-Up: Cool Down

Component Overview: Students will complete a Ticket Out the Door that assesses rewriting into vertex form by completing the square and identifying key features. This activity should be done independently to determine student mastery of the concept.



8.7.4 Wrap-Up: Cool Down

1. Rewrite the equation in vertex form by completing the square. Then, give the key features revealed by both standard form and vertex form.

$$y = -2x^2 + 12x + 1$$

$$y = -2(x - 3)^2 + 19$$

Key Features					
x-intercept		vertex	(3,19)	domain	R
x-intercept		increasing	$x < 3$	range	$y \leq 19$
y-intercept	(0,1)	decreasing	$x > 3$	opens	Down
left end behavior	$x \rightarrow -\infty, y \rightarrow -\infty$				
right end behavior	$x \rightarrow \infty, y \rightarrow -\infty$				



8.7.3 Guided Instruction utilizes a nearly identical table for identifying key features. Consider using the information from this activity to inform your instruction during question 12 of 8.8.2 Guided Instruction.



For question 1, students should only fill in the key features that are revealed. Students should not have to provide key features that cannot be found using either form.